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Compositions and methods for the treatment of ocular neovascularization

Abstract

The present invention relates to a non-aqueous ophthalmic composition comprising particles of a protein powder preparation suspended in a liquid vehicle comprising a semifluorinated alkane, wherein the particles of the protein powder preparation comprise an anti-VEGF protein selected from aflibercept or a sequence having at least 90% sequence identity to SEQ ID NO: 1. The composition is particularly suitable for the treatment of ocular neovascularization.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application is a U.S. National Stage application under 35 U.S.C. § 371 of International Application No. PCT/EP2020/053405, filed on Feb. 11, 2020, which claims priority to and the benefit of European Application No. 19156934.2, filed on Feb. 13, 2019, and European Application No. 19171487.2, filed on Apr. 29, 2019, the contents of each of which are hereby incorporated by reference in their entireties.

BACKGROUND OF THE INVENTION

(2) Neovascularization can occur in many eye diseases and its epidemiologic impact is significant. Ocular neovascularization is known from various parts of the eye, including the cornea, iris, retina, and choroid.

(3) Corneal neovascularization is characterized by the invasion of new blood vessels into the cornea from the limbus. It is caused by a disruption of the balance between angiogenic and antiangiogenic factors that preserves corneal transparency. Advanced stages, in which ingrown blood vessels reach the visual axis, can become permanently vision-threatening and, in patients with corneal grafts, may contribute to rejection.

(4) Several medical approaches, all of which are used off label, are available for treating corneal neovascularization. Recent studies involving monoclonal anti-VEGF antibodies have shown promising results for the reduction of corneal neovascularization. Topical and/or subconjunctival administration of bevacizumab or ranibizumab has demonstrated good short-term safety and efficacy although long-term data are lacking. Anti-VEGF therapy for corneal neovascularization is still considered experimental and off label, special consents are required, and insurance coverage may be denied. In case of cornea transplants, corneal neovascularization greatly elevates the risk of graft rejection and, ultimately, failure in patients undergoing corneal transplants.

(5) The following documents discuss the use of anti-VEGF proteins in the treatment of corneal neovascularization.

(6) U.S. Pat. No. 7,608,261 describes aqueous solutions of anti-VEGF protein that upon injection inhibit injury-induced corneal neovascularization

(7) WO2007/149334 describes the stability of aqueous ophthalmic formulations of anti-VEGF proteins.

(8) WO2016/208989 describes aqueous solutions of anti-VEGF proteins that are characterized by high stability, e.g. for long term storage.

(9) Park et al (Cornea. 2015; 34(10):1303-7) show that the topical administration of aqueous 0.1% aflibercept or 0.1% bevacizumab have comparable inhibitory effects on corneal neovascularization in rabbits.

(10) Sella et al. (Exp Eye Res. 2016; 146:224-32) compare the inhibitory effect on corneal neovascularization using topical administration of high-concentrated aqueous aflibercept and bevacizumab formulations in a rat model of chemical burn.

(11) WO2015/011199 describes compositions comprising an antigen-binding protein, and a liquid vehicle comprising a semifluorinated alkane.

SUMMARY OF THE INVENTION

(12) The present invention is in the field of ophthalmology. In particular, the present invention

relates to methods and compositions for the treatment of ocular neovascularization, especially corneal neovascularization.

Description

BRIEF DESCRIPTION OF THE FIGURES

- (1) FIG. 1: Inhibition of blood vessel growth (hemeangiogenesis) in suture-induced inflammatory neovascularization mouse model utilizing the anti-VEGF protein aflibercept. Afli/F6H8=protein powder preparation of aflibercept suspended in 1-perfluorohexyloctane (F6H8) at a protein concentration of 5 mg/ml; Afli=aflibercept aqueous solution in 10 mM sodium phosphate buffer, pH 6.2 at a protein concentration of 5 mg/ml; F6H8=vehicle (1-perfluorohexyloctane); Phosphate=control (aqueous 10 mM sodium phosphate buffer, pH 6.2).
- (2) FIG. 2: Inhibition of lymph vessel growth (lymphangiogenesis) in suture-induced inflammatory neovascularization mouse model utilizing the anti-VEGF protein aflibercept. Afli/F6H8=protein powder preparation of aflibercept suspended in 1-perfluorohexyloctane (F6H8) at a protein concentration of 5 mg/ml; Afli=aflibercept aqueous solution in 10 mM sodium phosphate buffer, pH 6.2 at a protein concentration of 5 mg/ml; F6H8=vehicle (1-perfluorohexyloctane); Phosphate=control (aqueous 10 mM sodium phosphate buffer, pH 6.2).
- (3) FIG. 3: Inhibition of blood vessel growth (hemeangiogenesis) in suture-induced inflammatory neovascularization mouse model utilizing the anti-VEGF protein bevacizumab. Be/F6H8=protein powder preparation of bevacizumab suspended in 1-perfluorohexyloctane (F6H8) at a protein concentration of 5 mg/ml; Bevacizumab=bevacizumab aqueous solution in 10 mM sodium phosphate buffer, pH 6.2 at a protein concentration of 5 mg/ml; F6H8=vehicle (1-perfluorohexyloctane); Phosphate=control (aqueous 10 mM sodium phosphate buffer, pH 6.2).
- (4) FIG. 4: Inhibition of lymph vessel growth (lymphangiogenesis) in suture-induced inflammatory neovascularization mouse model utilizing the anti-VEGF protein bevacizumab. Be/F6H8=protein powder preparation of bevacizumab suspended in 1-perfluorohexyloctane (F6H8) at a protein concentration of 5 mg/ml; Afli=bevacizumab aqueous solution in 10 mM sodium phosphate buffer, pH 6.2 at a protein concentration of 5 mg/ml; F6H8=vehicle (1-perfluorohexyloctane); Phosphate=control (aqueous 10 mM sodium phosphate buffer, pH 6.2).
- (5) FIG. 5 Inhibition of blood vessel growth and lymph vessel growth in the suture-induced inflammatory neovascularization mouse model utilizing the anti-VEGF protein aflibercept at an early timepoint, namely after 5 and 10 days after suture induction (dosage regimen: 3-times daily, each 3 μ l).
- (6) FIG. 6 Inhibition of blood vessel growth and lymph vessel growth in the suture-induced inflammatory neovascularization mouse model utilizing the anti-VEGF protein aflibercept utilizing a low dose application scheme (dosage regimen: 1-times daily; 3 μ l) at an early timepoint after 5 days after suture induction.
- (7) FIG. 7 Infiltration of inflammatory cells such as CD45+ cells and LYVE-1 macrophages at the early timepoint after 3 days after suture induction.

DETAILED DESCRIPTION OF THE INVENTION

- (8) The invention is based on the finding of the inventors that ophthalmic compositions comprising particles of a powder preparation suspended in a liquid vehicle comprising a semifluorinated alkane, wherein the particles comprise an anti-VEGF-protein, are suitable for the treatment of ocular neovascularization.
- (9) As such, in a first aspect, the invention relates to a non-aqueous ophthalmic composition comprising particles suspended in a liquid vehicle comprising a semifluorinated alkane, wherein the particles comprise an anti-VEGF protein. Preferably, said particles are particles of a protein powder preparation.

(10) In particular, the invention relates to a non-aqueous ophthalmic composition comprising particles of a protein powder preparation suspended in a liquid vehicle, wherein the particles of a protein powder preparation comprise an anti-VEGF protein and wherein the liquid vehicle comprises a semifluorinated alkane.

(11) In the context of the present invention VEGF refers to Vascular endothelial growth factor. Vascular endothelial growth factor is a signal protein produced by cells that stimulates the formation of blood vessels (angiogenesis)

(12) VEGF refers to a family of growth factors with several members. In mammals the family comprises VEGF-A to VEGF-D. Within the context of the present invention the term VEGF refers to any family member of the VEGF family.

(13) In the context of the present invention, the term anti-VEGF protein refers to proteins that neutralize VEGF activity. In some embodiments, said anti-VEGF protein is an antigen binding polypeptide or protein that binds to a VEGF antigen. Polypeptides and proteins in general represent polymers of amino acid units that linked to each other by peptide bonds. Since the size boundaries that are often used to differentiate between polypeptides and proteins are somewhat arbitrary, the two expressions for these molecules should—within the context of the present invention—not be understood as mutually exclusive: A polypeptide may also be referred to as a protein, and vice versa. Typically, the term “polypeptide” only refers to a single polymer chain, whereas the expression “protein” may also refer to two or more polypeptide chains that are linked to each other by non-covalent bonds.

(14) More specifically, and as used within the context of the present invention, antigen-binding polypeptides or proteins refer to full-length and whole antibodies (also known as immunoglobulins) in their monomer, or polymeric forms and any fragments, chains, domains or any modifications derived from a full-length antibody capable of specifically binding to an antigen. The antigen-binding polypeptides or proteins may belong to any of the IgG, IgA, IgD, IgE, or IgM immunoglobulin isotypes or classes. Fusion proteins comprising an antibody fragment capable of specifically binding to an antigen and antibody-drug conjugates are also within the definition of antigen-binding polypeptides or proteins as used herein.

(15) A full-length antibody is a Y-shaped glycoprotein comprising of a general structure with an Fc (fragment crystallisable) domain and a Fab (fragment antigen binding) domain. These are structurally composed from two heavy (H) chains and two light (L) chain polypeptide structures interlinked via disulfide bonds to form the Y-shaped structure. Each type of chain comprises a variable region (V) and a constant region (C); the heavy chain comprises a variable chain region (V.sub.H) and various constant regions (e.g. C.sub.H1, C.sub.H2, etc.) and the light chain comprises a variable chain region (V.sub.L) and a constant region (C.sub.L). The V regions may be further characterized into further sub-domains/regions, i.e. framework (FR) regions comprising more conserved amino acid residues and the hypervariable (HV) or complementarity determining regions (CDR) which comprise of regions of increased variability in terms of amino acid residues. The variable regions of the chains determine the binding specificity of the antibody and form the antigen-binding Fab domains of an antibody.

(16) As used herein anti-VEGF proteins include, but are not limited to, anti-VEGF antibodies and related molecules. In a preferred embodiment of the invention, the compositions comprise an anti-VEGF protein, wherein the anti-VEGF protein is selected from a monoclonal antibody, polyclonal antibody, an antibody fragment, a fusion protein comprising an antibody fragment, an antibody-drug conjugate, or any combination thereof.

(17) In a particularly preferred embodiment of the invention, the compositions comprise an anti-VEGF protein selected from a monoclonal antibody (mAb). A monoclonal antibody refers to an antibody obtained from a homogenous population of antibodies that are specific towards a single epitope or binding site on an antigen. Monoclonal antibodies may be produced using antibody engineering techniques known in the art, such as via hybridoma or recombinant DNA methods.

(18) Also within the scope of anti-VEGF proteins are antibody fragments binding to a VEGF-antigen. Antibody fragments of the invention include any region, chain, domain of an antibody, or any constructs or conjugates thereof that can interact and bind specifically to a VEGF-antigen, and may be monovalent, bivalent, or even multivalent with respect to binding capability. Such antibody fragments may be produced from methods known in the art, for example, dissection (e.g. by proteolysis) of a full-length native antibody, from protein synthesis, genetic engineering/DNA recombinant processes, chemical cross-linking or any combinations thereof. Antibody fragments are commonly derived from the combination of various domains or regions featured in variable V region of a full-length antibody.

(19) Preferably, the anti-VEGF protein of the present invention is selected from bevacizumab (Avastin), aflibercept (VEGF Trap-Eye; EYLEA®) and ziv-aflibercept (VEGF Trap; ZALTRAP®)

(20) Aflibercept

(21) Aflibercept (commercial name, EYLEA) is an anti-VEGF protein, namely a Fc-fusion polypeptide carrying extracellular domains of VEGF receptors, being used as decoy receptor to neutralize VEGF. Aflibercept is for treating patients suffering from Neovascular (Wet) Age-related Macular Degeneration (AMD), Macular Edema following Retinal Vein Occlusion (RVO), Diabetic Macular Edema (DME) and Diabetic Retinopathy (DR). The amino acid sequence of aflibercept (also known as VEGFR1 R2-FcAC1 (a)), as well as the nucleic acid sequence encoding the same, are set forth, e.g., in WO2012/097019.

(22) The Protein sequence (SEQ ID NO: 1) for aflibercept is:

(23) TABLE-US-00001 SDTGRPFVEMYSEIPEIIHMTGRELVIPCRVTSPNITVTLKKFPLDTL
IPDGKRIIWD SRKGFIISNATYKEIGLLICEATVNGHLYKTNYLTHRQI
NTIIDVVLSPSHGIELSVGEKLVLNCTARTELVNGIDFNWEYPSSKHQH
KKLVNRDLKTQSGSEMKKFLSILTIDGVTRSDQGLYTCAASSGLMTKKN
STFVRVHEKDKTHTCPPCPAPPELLGGPSVFLFPPKPKDTLMISRTPEVT
CVVVDVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVL
HQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSRDEL
TKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTTPVLDSDGSFFLY
SKLTVDKSRWQQGNVFCFSVMHEALHNHYTQKSLSLSPG

Ziv-Aflibercept

(24) Ziv-aflibercept contains the same protein (active drug) as aflibercept but is specifically formulated for injection as an intravenous infusion. Ziv-aflibercept is not intended for ophthalmic use, as the osmolarity of the ziv-aflibercept preparation is significantly higher than that of intravitreal aflibercept injection. However, intravitreal ziv-aflibercept has been used with success for multiple ocular conditions with acceptable safety profile.

(25) Bevacizumab

(26) Bevacizumab is a recombinant humanized monoclonal antibody that blocks angiogenesis by inhibiting vascular endothelial growth factor A (VEGF-A). Bevacizumab is a full-length IgG1 κ isotype antibody composed of two identical light chains (214 amino acid residues; SEQ ID NO: 2) and two heavy chains (453 residues; SEQ ID NO: 3) with a total molecular weight of 149 kDa. The two heavy chains are covalently coupled to each other through two inter-chain disulfide bonds, which is consistent with the structure of a human IgG1.

(27) The protein sequence of the "Bevacizumab light chain" (SEQ ID NO: 2)

(28) TABLE-US-00002

DIQMTQSPSSLSASVGDRVTITCSASQDISNYLNWYQQKPGKAPKVLIIY
FTSSLHSGVPSRFGSGSGTDFLTISLQPEDFATYYCQQYSTVPWTF
GQGTEKVEIKRTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQ
WKVDNALQSGNSQESVTEQDSKSTYSLSTLTLSKADYEKHKVYACEV
THQGLSSPVTKSFNRGEC

(29) The protein sequence of the "Bevacizumab heavy chain" (SEQ ID NO: 3) is:

EVQLVESGGGLVQPGGSLRLSCAASGYTFTNYGMNWVRQAPGKGLEWVG
WINTYTGEPTYAADFKRRFTFSLDTSKSTAYLQMNSLRAEDTAVYYCAK
YPHYYGSSHWYFDVWGQGTLVTVSSASTKGPSVFPLAPSSKSTSGGTAA
LGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPS
SSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCPPCPAPELLGGPS
VFLFPPKPKDTLMISRTPEVICVVDVSHEDPEVKFNWYVDGVEVHNAK
TKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTIS
KAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWESNGQ
PENNYKTTTPVLDSGDSFFLYSKLTVDKSRWQQGNVFSVMSVMHEALHNH
YTQKSLSLSPGK

(31) As understood herein, anti-VEGF antibodies may be chimeric, humanized or human. Chimeric monoclonal antibodies, for example, refer to hybrid monoclonal antibodies comprising domains or regions of the heavy or light chains derived from antibody sequences from more than one species, for example from murine and human antibody sequences. Humanized monoclonal antibodies refer to those that are predominantly structurally derived from human antibody sequences, generally with a contribution of at least 85-95% human-derived sequences, whereas the term human refers to those that are derived solely from human germline antibody sequences. In a preferred embodiment, the compositions comprise of an antigen-binding polypeptide or protein selected from a monoclonal antibody, wherein the monoclonal antibody is a chimeric, humanized, or human antibody.

(32) The liquid vehicle comprises a semifluorinated alkane. Semifluorinated alkanes provide a number of advantages from the pharmaceutical perspective. They are substantially non-toxic and are found to be well-tolerated by various types of human and animal tissue when administered topically or parenterally. In addition, they are chemically inert and are generally compatible with active and inactive ingredients in pharmaceutical formulations. Semifluorinated alkanes, when acting as vehicles for compounds that are not soluble or poorly soluble (such as antigen-binding proteins or polypeptides), form dispersions or suspensions with very useful physical or pharmaceutical properties, i.e. with little or no tendency to form solid, non-dispersible sediments.

(33) Semifluorinated alkanes are linear or branched alkanes some of whose hydrogen atoms have been replaced by fluorine. In the semifluorinated alkanes (SFAs) used in the present invention, one linear non-fluorinated hydrocarbon segment and one linear perfluorinated hydrocarbon segment are present. These compounds thus follow the general formula $F(CF_2)_n(CH_2)_mH$. According to the present invention, n is selected from the range of 3 to 12, and m is selected from the range of 3 to 12, preferably n is selected from the range of 4 to 8, and m is selected from the range of 4 to 10.

(34) A nomenclature which is frequently used for semifluorinated alkanes designates a perfluorinated hydrocarbon segment as RF and a non-fluorinated segment as RH. Alternatively, the compounds may be referred to as F_nH_m and F_nH_m , respectively, wherein F means a perfluorinated hydrocarbon segment, H means a non-fluorinated segment, and n and m define the number of carbon atoms of the respective segment. For example, F_3H_3 is used for perfluoropropylpropane, $F(CF_2)_3(CH_2)_3H$. Moreover, this type of nomenclature is usually used for compounds having linear segments. Therefore, unless otherwise indicated, it should be assumed that F_3H_3 means 1-perfluoropropylpropane, rather than 2-perfluoropropylpropane, 1-perfluoroisopropylpropane or 2-perfluoroisopropylpropane.

(35) Preferred semifluorinated alkanes include in particular the compounds F_4H_5 , F_4H_6 , F_4H_8 , F_6H_4 , F_6H_6 , F_6H_8 , and F_6H_{10} . Particularly preferred for carrying out the invention are F_4H_5 , F_4H_6 , F_6H_6 and F_6H_8 . In another particularly preferred embodiment, the composition of the invention comprises F_6H_8 .

(36) Optionally, the composition may comprise more than one SFA. It may be useful to combine SFAs, for example, in order to achieve a particular target property such as a certain density or

viscosity. If a mixture of SFAs is used, it is furthermore preferred that the mixture comprises at least one of F4H5, F4H6, F6H4, F6H6, F6H8, and F6H10, and in particular one of F4H5, F4H6, F6H6 and F6H8. In another embodiment, the mixture comprises at least two members selected from F4H5, F4H6, F6H4, F6H6, F6H8, and F6H10, and in particular at least two members selected from F4H5, F6H6 and F6H8.

(37) Liquid SFAs are chemically and physiologically inert, colourless and stable. Their typical densities range from 1.1 to 1.7 g/cm³, and their surface tension may be as low as 19 mN/m. SFAs of the RFRH type are insoluble in water but also somewhat amphiphilic, with increasing lipophilicity correlating with an increasing size of the non-fluorinated segment.

(38) The composition comprises a liquid vehicle, wherein the semifluorinated alkane is present at a concentration of at least 85% by weight of the composition (wt %), preferably the semifluorinated alkane is present at a concentration of about 85 to 99% by weight of the composition.

(39) It has been found by the inventors that the presence of a semifluorinated alkane in a liquid vehicle or in particular as a liquid vehicle in a composition comprising an anti-VEGF protein has a remarkable stabilizing effect on anti-VEGF proteins. In particular, compositions comprising semifluorinated alkane as a liquid vehicle are capable of substantially preventing or inhibiting their aggregation and reducing chemical degradation over a substantial period of time, at room temperature and even at higher temperatures such as 40° C., without loss of biological activity. Furthermore, it has been found that a composition comprising an anti-VEGF protein and a liquid vehicle comprising a semifluorinated alkane shows anti-neovascularization activity when topically applied to the surface of the eye.

(40) It has also been found that anti-VEGF protein dispersions and suspensions in semifluorinated alkanes exhibit a remarkable degree of physical stability. The occurrence of flotation or sedimentation takes place slowly, leaving sufficient time for the withdrawal of a dose after gentle shaking or swirling of the container (e.g. a vial) with the dispersion or suspension. The anti-VEGF protein particles in semifluorinated alkane appear to largely retain their original particle size distribution and are readily redispersible; poorly re-dispersible aggregates do not appear to be formed. Importantly, this provides for a higher level of dosing accuracy in terms of precision and reproducibility.

(41) In contrast, suspensions or dispersions in other chemically inert vehicles tend to be unstable, leading to formation of dense and poorly redispersible aggregates, and making precise dosing challenging, or in some cases, impossible, such as leading to the clogging of fine-gauged needles typically used for subcutaneous injections. Aggregated protein particles also present a high risk towards triggering adverse immunogenic reactions.

(42) The term protein powder preparation refers to a protein composition in powdered form, preferably it is in form of dry solid particles that comprise at least an anti-VEGF protein as defined above. The particles of the protein powder preparations may be obtained by lyophilization or by spray-drying from an aqueous solution comprising an anti-VEGF protein

(43) In a particular aspect, the invention therefore relates to an ophthalmic composition comprising particles of a protein powder preparation suspended in a liquid vehicle which comprises a semifluorinated alkane of the formula RFRH, wherein RF is a linear perfluorinated hydrocarbon segment with 4 to 8 carbon atoms, and wherein RH is a linear alkyl group with 4 to 10 carbons. Moreover, said particles of the protein powder preparation comprise an anti-VEGF protein. The particles of the protein powder preparation comprising an anti-VEGF protein are incorporated in the ophthalmic composition such as to form a suspension in the liquid vehicle.

(44) The term liquid vehicle comprising a semifluorinated alkane refers to a liquid composition comprising at least one semifluorinated alkane. The liquid vehicle may comprise further compounds. Preferably, the liquid vehicle comprises one or more semifluorinated alkanes. More preferably, the liquid vehicle consists of at least one semifluorinated alkane or a mixture of semifluorinated alkanes.

(45) Accordingly, in one embodiment, the invention relates to an ophthalmic composition comprising particles of a protein powder preparation suspended in a liquid vehicle, wherein the liquid vehicle consists of a semifluorinated alkane or a mixture of semifluorinated alkanes of formula $F(CF_2)_n(CH_2)_mH$, wherein n is an integer from 4 to 8 and m is an integer from 4 to 10 and wherein the particles of the protein powder preparation comprise an anti-VEGF protein. In a particular embodiment, the liquid vehicle consists of a semifluorinated alkane of formula $F(CF_2)_n(CH_2)_mH$, wherein n is an integer from 4 to 8 and m is an integer from 4 to 10. In a preferred embodiment, said semifluorinated alkane or said mixture of semifluorinated alkanes is selected from at least one of F4H5, F4H6, F6H4, F6H6, F6H8, and F6H10, and in particular one of F4H5, F4H6, F6H6 and F6H8. In a specific embodiment the liquid vehicle consists of F6H8.

(46) The anti-VEGF protein may be any suitable anti-VEGF protein as defined above. Preferably, the anti-VEGF protein is a fusion protein binding to VEGF. In one embodiment, the anti-VEGF protein comprises a sequence having at least 90% sequence identity to SEQ ID NO: 1. In one embodiment the anti-VEGF protein consists of a sequence having at least 90% sequence identity to SEQ ID NO: 1. In a further embodiment the, the anti-VEGF protein comprises a sequence having 100% sequence identity to SEQ ID NO. 1. In a further embodiment, the anti-VEGF protein consists of a sequence having 100% sequence identity to SEQ ID NO. 1. In a preferred embodiment the anti-VEGF protein is aflibercept or ziv-aflibercept.

(47) In a further embodiment, the anti-VEGF protein is an antibody binding to VEGF. In one embodiment the anti-VEGF protein comprises a sequence having at least 90% sequence identity to SEQ ID NO: 2 or SEQ ID NO: 3. In a further embodiment, the anti-VEGF protein comprises protein sequences having at least 90% sequence identity to SEQ ID NO: 2 and SEQ ID NO: 3. In a particular embodiment, the anti-VEGF protein comprises at least two polypeptides, wherein each polypeptide consists of a sequence having at least 90% sequence identity to SEQ ID NO: 2 or of a sequence having at least 90% sequence identity to SEQ ID NO: 3. In some embodiments, the anti-VEGF protein consists of polypeptides comprising sequences having at 100% sequence identity to SEQ ID NO: 2 and SEQ ID NO: 3. In a further embodiment, the anti-VEGF protein consists of polypeptides consisting of sequences having at 100% sequence identity to SEQ ID NO: 2 and SEQ ID NO: 3. In a particular embodiment, the anti VEGF-protein is bevacizumab (avastin).

(48) In some embodiments of the invention, the concentration of the anti-VEGF protein in the ophthalmic suspension of the present invention is up to about 60 mg/ml. In some embodiments, the concentration of the anti-VEGF protein in the ophthalmic suspension is in between about 1 to 60 mg/ml, preferably in between about 1 to 25 mg/ml, more probably between about 1 to 10 mg/ml, even more preferably the concentration of the anti-VEGF protein is about 5 mg/ml. In particular embodiments, the concentration of the anti-VEGF protein in the suspension is 60 mg/ml, 50 mg/ml, 40 mg/ml, 30 mg/ml, 20 mg/ml, 15 mg/ml, 10 mg/ml, 7.5 mg/ml, 5 mg/ml, 4 mg/ml, 3 mg/ml, 2.5 mg/ml, 2 mg/ml or 1 mg/ml.

(49) The ophthalmic composition comprising particles suspended in a liquid vehicle may comprise any suitable amount of a protein powder preparation comprising an anti-VEGF-protein. The inventors found that the composition preferably comprises about up to 35% weight (wt %) of the protein powder preparation comprising an anti-VEGF-protein. As such, in one embodiment, the composition comprises up to about 35% by weight (wt %) of the protein powder preparation comprising an anti-VEGF-protein. In some embodiments the composition comprises between 0.1 and 16% by weight (wt%) of the protein powder preparation comprising an anti-VEGF-protein. In a preferred embodiment, the composition comprises between 0.5% and 8% by weight (wt %) of the protein powder preparation comprising an anti-VEGF-protein. In some embodiments, the composition comprises about 0.1%, 0.25%, 0.5%, 1.0%, 1.5%, 2.0%, 2.5%, 3.0%, 3.5%, 4.0%, 4.5%, 5.0%, 6.0%, 7.0%, 7.5%, 8.0%, 9.0%, 10.0%, 15.0%, 17.5%, 20.0%, 25%, 30%, 3 or 35.0% by weight (wt %) of the protein powder preparation comprising an anti-VEGF-protein. In a

preferred embodiment, the composition comprises 1.0 to 4.0%, preferably about 1.2 or 3.8% by weight (wt %) of the protein powder preparation comprising an anti-VEGF-protein.

(50) As noted before the particles of the protein powder preparation comprising an anti-VEGF protein may be obtained by lyophilization and/or spray drying. In some embodiments the particles of the protein powder preparation comprising an anti-VEGF protein are obtained by spray drying or lyophilization of an aqueous solution comprising an anti-VEGF protein. Said aqueous solution may further comprise protein stabilizing agents. Alternatively, or in addition, said aqueous solutions may comprise a buffering agent.

(51) Suitable protein stabilizing agents and buffering agents are known to the skilled person. Preferably, said one or more protein stabilizing agents are selected from polyols, sugars (i.e. sucrose or trehalose), amino acids, amines and salting out salts, polymers, surfactants, and arginine.

(52) Said buffering agent may be any pharmacologically acceptable buffering agent A preferred buffering agent is sodium phosphate.

(53) In some embodiments, the particles of the protein powder preparation comprise an anti-VEGF protein and one or more protein stabilizing agents selected from polyols, sugars (i.e. sucrose or trehalose), amino acids, amines and salting out salts, polymers, surfactants, and arginine.

(54) In a preferred embodiment of the invention, the particles of the protein powder preparation are obtained by lyophilization of an aqueous solution comprising an anti-VEGF protein, one or more protein stabilizing agents as defined above and a buffering agent. In a further preferred embodiment of the invention, the particles of the protein powder preparation are obtained by spray drying of an aqueous solution comprising an anti-VEGF protein, one or more protein stabilizing agents as defined above and a buffering agent. Preferably, said aqueous solution has a physiologically and pharmacologically acceptable pH. More preferably, said solution has an ophthalmologically acceptable pH.

(55) Accordingly, in some embodiments, the invention relates to an ophthalmic composition comprising particles of a protein powder preparation suspended in a liquid vehicle comprising a semifluorinated alkane, wherein the particles of the protein powder preparation comprise an anti-VEGF protein, one or more protein stabilizing agents and optionally a buffering agent.

(56) In a preferred embodiment said protein stabilizing agent is selected from polyols, sugars (i.e. sucrose or trehalose), amino acids, amines and salting out salts, polymers, surfactants, and arginine.

(57) In a particular embodiment, the protein stabilizing agent is selected from sucrose, trehalose and polysorbate. In a preferred embodiment, said protein stabilizing agent is trehalose.

(58) The total concentration of said particles in the ophthalmic composition according to the invention is up to 400 mg/ml. Preferably, the concentration of said particles in the ophthalmic composition is in between about 10 to 400 mg/ml, preferably between about 10 to 200 mg/ml, more preferably between 10 to 100 mg/ml, even more preferably between about 10 to 60 mg/ml.

(59) Accordingly, in one embodiment, the invention relates to a composition as defined above, wherein the total solid content of the composition comprising the protein powder preparation suspended in the liquid vehicle comprising the semifluorinated alkane is in between about 10 to 400 mg/ml, preferably between about 10 to 200 mg/ml, more preferably between 10 to 100 mg/ml, even more preferably between about 10 to 60 mg/ml.

(60) The invention further relates to a composition as defined above, wherein the size of 90% of the suspended particles of the protein powder preparation is between 1 to 100 μm , preferably it is between 1 to 50 μm , more preferably it is between 1 to 30 μm , even more preferably it is between 1 to 20 μm , as measured by laser diffraction.

(61) It is a particular property of the invention that the anti-VEGF protein in the composition as defined above maintains its activity over storage, especially over a long period of time and even at elevated temperatures. The inventors found, that a composition as defined above, is able to stabilize an anti-VEGF protein in the absence of added preservative agents. As such, the invention relates, in some embodiments, to a composition as defined above, wherein the composition is free of a

preservative.

(62) In a particular embodiment, the invention relates to a composition as defined above wherein the anti-VEGF protein comprised in the suspension retains 90% of its activity when stored at 2-8° C. for up to 3 months or wherein the anti-VEGF protein comprised in the suspension retains 90% of its activity when stored at 25° C. and 60% humidity for up to 3 months. (see Example 2(c) stability)

(63) In a further embodiment, the invention relates to a composition as defined above, wherein 90% of the suspended (anti-VEGF protein containing) particles retain their initial particle size when stored at 2-8° C. for up to 3 months or wherein 90% of the suspended (anti-VEGF protein containing) particles retain their initial particle size when stored at 25° C. and 60% humidity for up to 3 months.

(64) In a preferred embodiment, the ophthalmic composition comprises anti-VEGF containing particles of a protein powder preparation suspended in a vehicle comprising F6H8, a protein stabilizing agent and a buffering agent.

(65) In a further preferred embodiment, the ophthalmic composition comprises aflibercept containing particles of a protein powder preparation suspended in a vehicle comprising F6H8, sucrose and a buffering agent, wherein the concentration of aflibercept in the composition is about 5 mg/ml.

(66) In a further preferred embodiment, the ophthalmic composition comprises bevacizumab containing particles of a protein powder preparation suspended in a vehicle comprising F6H8, trehalose and a buffering agent, wherein the concentration of bevacizumab in the composition is about 5 mg/ml.

(67) The compositions of the invention are particularly suitable for medical uses, in particular ophthalmic uses. As such, in one embodiment, the invention relates to an ophthalmic composition as defined above in all embodiments for use as a medicament. In a further embodiment, the invention relates to an ophthalmic composition as defined above in all embodiments for use in the manufacture of a medicament.

(68) The inventors surprisingly found, that a composition as defined above in all embodiments, i.e. a composition comprising a suspension of particles of a protein powder preparation (comprising an anti-VEGF protein) in a liquid vehicle comprising an SFA is as effective in the treatment of ocular neovascularization as an aqueous composition (solution) comprising said anti-VEGF protein.

(69) As such, in a particular aspect, the invention relates to an ophthalmic composition comprising particles of a protein powder preparation suspended in a liquid vehicle comprising a semifluorinated alkane for use in a method for treatment and/or prevention of ocular neovascularization, wherein the protein powder preparation comprises an anti-VEGF protein. In a particular embodiment, the invention relates to an ophthalmic composition comprising particles of a protein powder preparation suspended in a liquid vehicle comprising a semifluorinated alkane for use the manufacture of a medicament for treatment and/or prevention of ocular neovascularization, wherein the protein powder preparation comprises an anti-VEGF protein.

(70) In some embodiments, the invention relates to a composition as defined above in all embodiments for use in a method of treatment and/or prevention of ocular neovascularization. The invention further relates to the use of a composition as defined above in a method for treatment and/or prevention of ocular neovascularization.

(71) The inventors found that compositions as defined above in all embodiments show an inhibition of blood vessel growth comparable or better than aqueous compositions comprising the same anti-VEGF protein. Accordingly, in a particular embodiment, the invention relates to a composition as defined above in all embodiments for use in a method of the inhibition of growth of blood vessels and/or lymphatic vessels. In some embodiments the invention relates to a composition as defined above in all embodiments for use in the manufacture of a medicament for use in a method of the inhibition of growth of blood vessels and/or lymphatic vessels.

(72) In a particular embodiment of the invention, the invention relates to a composition as defined above in all embodiments for use in a method of the simultaneous inhibition of growth of blood vessels and lymphatic vessels. In some embodiments the invention relates to a composition as defined above in all embodiments for use in the manufacture of a medicament for use in a method of the simultaneous inhibition of growth of blood vessels and lymphatic vessels.

(73) The compositions according to the invention are suitable to be used for the treatment of any type of ocular neovascularization and particularly suitable for the treatment and/or prevention of corneal neovascularization, in particular for the treatment, prevention and/or control of corneal angiogenesis and/or corneal lymphangiogenesis.

(74) Accordingly, in one embodiment, the invention relates to a composition as defined above in all embodiments for use in a method of treatment and/or prevention of corneal neovascularization. In a further embodiment, the invention relates to a method as defined above in all embodiments for use in the manufacture of a medicament for the treatment and/or prevention or corneal neovascularization.

(75) In a particular embodiment, the invention relates to a composition as defined above in all embodiments for use in a method of prevention and/or treatment of corneal angiogenesis and/or corneal lymphangiogenesis. In a further embodiment, the invention relates to the use of said compositions in a method of manufacture of a medicament for use in a method of prevention and/or treatment of corneal angiogenesis and/or corneal lymphangiogenesis.

(76) The invention also relates to the use of compositions as defined above in all embodiments in a method of treatment or prevention of ocular neovascularization as defined above. In such a use, the composition is preferably applied to the eye or an ophthalmic tissue. More preferably, the composition as defined above is applied topically to the eye or an ophthalmic tissue.

(77) As such, the composition and the application are particularly suitable for subjects or patients, which may be affected by neovascularization. Such subjects include patients about to or having received ophthalmic surgery, in particular corneal surgery, especially corneal transplantations.

(78) The invention further relates to a method for treatment and/or prevention of ocular neovascularization comprising administering a composition as defined above in all embodiments to the eye or an ophthalmic tissue of a subject. In a particular embodiment, said method is a method for treatment and/or prevention of corneal neovascularization, especially treatment or prevention of corneal angiogenesis and/or corneal lymphangiogenesis.

(79) The invention further relates to a method for inhibiting blood vessel and/or lymphatic vessel growth comprising administering a composition as defined above in all embodiments to the eye or an ophthalmic tissue of a subject.

(80) In either method of the invention, the composition as defined herein, is preferably applied topically to the eye or ophthalmic tissue of a subject.

(81) Said subject preferably is a vertebrate, more preferably a mammal. In some embodiments, the invention relates to a method as defined above, in all embodiments, wherein the subject is a mammal. In a particular embodiment, the subject is a human.

(82) In some embodiments, the invention relates to a method as defined above, wherein the subject is at risk of being affected by or is affected by ocular neovascularization. In a particular embodiment, said subject has received or is receiving ophthalmic surgery, in particular corneal surgery. In a specific embodiment, the subject is receiving or has received corneal transplantation surgery.

(83) In a further aspect the invention relates to a kit comprising an ophthalmic composition as defined herein in all embodiments and a container adapted for administration to the eye. In a preferred embodiment the container is adapted for topical application.

(84) Preferably, said kit further comprises instructions for use.

(85) The invention further relates in particular to the following numbered items: 1. A non-aqueous ophthalmic composition comprising particles suspended in a liquid vehicle comprising a

semifluorinated alkane, wherein the particles comprise an anti-VEGF protein. 2. A non-aqueous ophthalmic composition according to item 1 comprising particles of a protein powder preparation suspended in a liquid vehicle comprising a semifluorinated alkane, wherein the particles of the powder preparation comprise an anti-VEGF protein. 3. The composition of item 1 or 2, wherein the anti-VEGF protein comprises a sequence having at least 90% sequence identity to SEQ ID NO: 1. 4. The composition of item 1 to 3, wherein the anti-VEGF protein is selected from aflibercept and ziv-aflibercept. 5. The composition of item 1 or 2, wherein the anti-VEGF protein comprises a sequence having at least 90% sequence identity to SEQ ID NO: 2 and SEQ ID NO: 3. 6. The composition of item 1, 2 or 5, wherein the anti-VEGF protein is bevacizumab. 7. The composition according to any preceding item, wherein the particles are obtained by spray-drying or lyophilization of an aqueous anti-VEGF protein solution. 8. The composition according to item 7, wherein the concentration of the anti-VEGF protein in the aqueous anti-VEGF protein solution is in between about 1 to 60 mg/ml, preferably in between about 1 to 25 mg/ml, more preferably between about 1 to 10 mg/ml, even more preferably the concentration of the anti-VEGF protein is about 5 mg/ml. 9. The composition according to items 6 or 7, wherein the aqueous anti-VEGF protein solution comprises one or more protein stabilizing agents selected from polyols, sugars (i.e. sucrose or trehalose), amino acids, amines and salting out salts, polymers, surfactants, and arginine. 10. The composition according to any preceding item, wherein the particles of the protein powder preparation comprise one or more protein stabilizing agents selected from polyols, sugars (i.e. sucrose or trehalose), amino acids, amines and salting out salts, polymers, surfactants, and arginine suspended in the liquid vehicle. 11. The composition according to any preceding item, wherein the particles of the protein powder preparation comprise a protein stabilizing agent selected from sucrose, trehalose and polysorbate. 12. The composition according to any preceding item, wherein the concentration of the anti-VEGF protein in the composition is in between about 1 to 60 mg/ml, preferably in between about 1 to 25 mg/ml, more preferably between about 1 to 10 mg/ml, even more preferably the concentration of the anti-VEGF protein is about 5 mg/ml. 13. The composition according to any preceding item, wherein the total solid content of the composition is in between about 10 to 400 mg/ml, preferably between about 10 to 200 mg/ml, more preferably between 10 to 100 mg/ml, even more preferably between about 10 to 60 mg/ml. 14. The composition according to any preceding item, wherein the size of 90% of the suspended particles of protein powder preparation is between 1 to 100 μm , preferably it is between 1 to 50 μm , more preferably it is between 1 to 30 μm , even more preferably it is between 1 to 20 μm , as measured by laser diffraction. 15. The composition according to any preceding item, wherein the semifluorinated alkane is of formula $\text{F}(\text{CF}_2)_n\text{—}(\text{CH}_2)_m\text{H}$, wherein n is an integer from 4 to 8 and m is an integer from 4 to 10. 16. The composition according to any preceding item, wherein the semifluorinated alkane is selected from F4H4, F4H5, F4H6, F6H4, F6H6, F6H8, F8H8. 17. The composition according to any preceding item, wherein the semifluorinated alkane is selected from 1-perfluorobutylpentane (F4H5) and 1-perfluorohexyloctane (F6H8), preferably the semifluorinated alkane is 1-perfluorohexyloctane (F6H8). 18. The composition according to any preceding item, wherein the semifluorinated alkane is present at a concentration of at least 85% by weight of the composition (wt %), preferably the semifluorinated alkane is present at a concentration of about 85 to 99% by weight of the composition. 19. The composition according to any preceding item, wherein the composition is free of a preservative. 20. The composition according to any preceding item, wherein the anti-VEGF protein comprised in the suspended particles retains 90% of its activity when the composition is stored at 2-8° C. for up to 3 months or wherein the anti-VEGF protein comprised in the suspended particles retains 90% of its activity when the composition is stored at 25° C. and 60% humidity for up to 3 months. 21. The composition according to any preceding item, wherein 90% of the suspended particles retain their initial particle size when the composition is stored at 2-8° C. for up to 3 months or wherein 90% of the suspended particles retain their initial particle size when the composition is stored at 25° C. and

60% humidity for up to 3 months. 22. The composition according to any of the preceding items, wherein the particles of the protein powder preparation are suspended in a vehicle comprising F6H8, sucrose and a buffering agent and wherein the composition comprises 5 mg/ml aflibercept 23. The composition according to any of the preceding items, wherein the particles of the protein powder preparation are suspended in a vehicle comprising F6H8, trehalose and a buffering agent and wherein the composition comprises 5 mg/ml bevacizumab 24. The composition according to any preceding items, for use as a medicament. 25. The composition for use according to item 24, in the prevention and/or treatment of ocular neovascularization. 26. The composition for use according to 24 to 25 in the inhibition of growth of blood vessels and/or lymphatic vessels. 27. The composition for use according to 24 to 26 in the simultaneous inhibition of growth of blood vessels and lymphatic vessels 28. The composition for use according to items 24 to 27, in the prevention and/or treatment of corneal neovascularization and/or corneal angiogenesis 29. The composition for use according to items 24 to 28, in the prevention and/or treatment of corneal angiogenesis and/or corneal lymphangiogenesis 30. The composition for use according to items 24 to 29, wherein the composition is topically administered to the eye or an ophthalmic tissue. 31. The composition for use according to 24 to 30, wherein the administration of the composition to a subject undergoing corneal transplantation. 32. A kit comprising an ophthalmic composition as defined in any of the items 1 to 23, a container for holding said composition adapted for administration to the eye or an ophthalmic tissue. 33. The kit of item 32, wherein the container is administered for topical administration. 34. A method for treating ocular neovascularization, comprising administering a composition according to any of the items 1 to 23 to the eye or an ophthalmic tissue. 35. The method according to any of the items 34, wherein the method is effective in inhibiting blood vessel and/or lymphatic vessel growth. 36. Use of a composition as defined in any of the items 1 to 23 for manufacture of a medicament for use in the treatment of ocular neovascularization. 37. A method for fabrication of a stable non-aqueous ophthalmic anti-VEGF-protein containing composition as defined in any of the items 1 to 23, comprising the steps of (a) subjecting an aqueous solution comprising an anti-VEGF protein, and optionally a protein stabilizing agent and/or a buffering agent, to spray-drying or lyophilization to form particles of a protein powder preparation (b) suspending the particles of said protein powder preparation in a liquid vehicle comprising a semifluorinated alkane to form the anti-VEGF protein containing composition

EXAMPLES

Example 1: Preparation of the Ophthalmic Compositions

(86) Starting from the aqueous commercial protein raw material of aflibercept (Zaltrap; Sanofi Genzyme; comprising 25 mg/ml aflibercept and additionally sucrose as protein stabilizer) and bevacizumab (Avastin; Roche; comprising 25 mg/ml bevacizumab and furthermore trehalose as protein stabilizer) the ophthalmic suspensions of the present invention were prepared with the following steps: (a) dilution of the aqueous commercial protein raw material to the desired concentration, (b) protein powder preparation (i.e. via spray-drying or lyophilization) of the diluted aqueous protein raw material of (a) and (c) preparation of the ophthalmic composition by suspending the protein powder preparations (b) in a liquid vehicle comprising a semifluorinated alkane (i.e. F6H8) ad a) The concentrated aqueous aflibercept (25 mg/ml) and bevacizumab (25 mg/ml) protein solutions were diluted with 5 mM sodium phosphate buffer, pH 6.2, to obtain a total solid content of 4-5% (w/v). ad b) The protein powder preparations comprising the anti-VEGF proteins (aflibercepts, bevacizumab) were e.g. prepared by spray-drying. Herein, the spray-drying was conducted using a Büchi B290 equipped with a high efficiency cyclone under the following conditions: Nozzle diameter 0.7 mm, drying air flow rate 35 m³/h, atomizing air flow rate 414 L/h. Obtained were protein powder preparations comprising aflibercept or bevacizumab. ad c) The ophthalmic SFA-containing compositions were prepared by suspending the appropriate amount of protein powder preparations (i.e. ~50 mg of a aflibercept or ~15 mg of bevacizumab containing protein power preparation) in F6H8 by vortexing, followed by 20 min sonication in an ultrasound

bath filled with ice water and intermittent vortexing (every 5 min) to receive 5 mg/ml aflibercept suspended in F6H8 (further comprising sucrose and buffering salts as protein stabilizer) and 5 mg/ml bevacizumab suspended in F6H8 (further comprising trehalose and buffering salts as protein stabilizer).

(87) As reference material aqueous ophthalmic solutions of aflibercept and bevacizumab were prepared by diluting the 25 mg/ml aqueous commercial protein raw material in sterile 10 mM sodium phosphate buffer (pH 6.2) to the desired final protein concentration of 5 mg/ml for aflibercept and bevacizumab.

Example 2: Analysis of the Ophthalmic Compositions

(88) (a) Reconstituted water-based protein powder preparation SEC-MALS (Size Exclusion Chromatography-Multi Angle Light Scattering) and ELISA was carried out on the aqueous commercial protein raw material (RM) of aflibercept and bevacizumab and the reconstituted protein powder preparation of aflibercept and bevacizumab to ensure that the manufacturing process did not change the protein. Herein, the reconstituted protein powder preparations (REP) of aflibercept and bevacizumab were obtained by extracting said anti-VEGF proteins from the non-aqueous SFA-based ophthalmic suspensions into water. In the extraction procedure equal volumes of water were added followed by gently shaking of the biphasic samples until the organic (semifluorinated alkane) phase becomes clear and all the anti-VEGF protein has moved into the aqueous phase. This aqueous layer (REP) was then transferred into a fresh vial and analysed for protein fragmentation/aggregation via SEC-MALS and for protein activity via ELISA. Protein aggregation and fragmentation

(89) Protein aggregation and fragmentation can be detected by first separating the fractions followed by determination of their molecular weight, i.e. SEC-MALS analysis.

(90) SEC-MALS analysis revealed that the chromatograms of the aqueous commercial protein raw material (RM) and the reconstituted protein powder preparations (REP) of both aflibercept and bevacizumab overlap very well and did not show an increase in aggregated or fragmented species. (b) Protein Activity

(91) ELISA was used to evaluate the activity of both proteins which involves the binding to VEGF, thereby inhibiting angiogenesis. The test is based on sandwich-type ELISA using a microtiter plate coated with recombinant human VEGF-A. Horseradish peroxidase (HRP)-conjugated anti-human IgG monoclonal antibody, which bind to the Fc region of antibodies, is then employed to quantify the bound Aflibercept or Bevacizumab. Both assays were performed using commercial kits from ImmunoGuide according to the manufacturer's instructions utilizing aqueous commercial protein raw material (RM) and reconstituted protein powder preparations (REP).

(92) The corresponding ELISA analyses also showed no significant difference in the binding activity between the RM and REP of both aflibercept and bevacizumab, therefore, the process of powder production as well as the subsequent preparation of the ophthalmic suspensions did not affect the activity of both anti-VEGF proteins (aflibercept, bevacizumab). (c) Stability

(93) Samples of the ophthalmic suspension of protein powder preparations of aflibercept and bevacizumab in F6H8 were prepared and placed on stability for three months. Two replicas were tested at each timepoint, initial, one month and 3 months. For the analysis of the stability samples, the anti-VEGF protein was extracted from the semifluorinated alkane-based ophthalmic suspension into the aqueous environment as described above.

(94) This stability study revealed that ophthalmic suspension of protein powder preparations of aflibercept (Table 1) and bevacizumab (Table 2) in F6H8 are stable in respect to assay, activity and aggregation when stored at 2-8° C. and 25° C./60% RH.

(95) TABLE-US-00004 TABLE 1 Stability study results aflibercept Temperature Test Method Initial 1 month 3 months 2-8° C. Assay UV - Absorption 4.01 mg/ml 4.11 mg/ml 4.28 mg/ml Activity ELISA 103.6% n. t. 90.8% Aggregation SEC-MALS <LOD 1.1% 0.25% 25° C./65% RH Assay UV - Absorption 4.01 mg/ml 4.19 mg/ml 4.25 mg/ml Activity ELISA 103.6% n. t.

96.5% Aggregation SEC-MALS <LOD 1.8% 0.5%

(96) TABLE-US-00005 TABLE 2 Stability study results bevacizumab Temperature Test Method
Initial 1 month 3 months 2-8° C. Assay UV - Absorption 4.28 mg/ml 4.23 mg/ml 4.42 mg/ml
Activity ELISA 91.5% 99.3% 94.7% Aggregation SEC-MALS 2.7% 3.2% 7.7% 25° C./65%
RH Assay UV - Absorption 4.28 mg/ml 4.20 mg/ml 4.33 mg/ml Activity ELISA 91.5% 92.1%
85.3% Aggregation SEC-MALS 2.7% 4.0% 18.2%

Example 3: Suture-Induced Inflammatory Corneal Neovascularization Assay

(97) Animals (BALB/c mice) were anesthetized with an intramuscular injection of ketamine (8 mg/kg) and xylazine (0.1 mL/kg) followed by placement of three 11-0 nylon sutures intrastromally with two stromal incursions extending over 120° of corneal circumference each. The outer point of suture placement was chosen near the limbus, and the inner suture point was chosen near the corneal centre equidistant from the limbus, to obtain standardized angiogenic responses. Sutures were left in place for the duration of the experiment.

(98) Following the surgical procedure, the mice were treated for 14 days, three times daily with 3 µl of control (buffer), ophthalmic compositions comprising protein powder preparation of bevacizumab and aflibercept suspended in F6H8 or blanks (10 mice per sample).

(99) The tested ophthalmic anti-VEGF protein compositions included:

(100) TABLE-US-00006 further Anti-VEGF Protein composition protein concentration vehicle composition components aflibercept 5 mg/ml 10 mM sodium aqueous sucrose phosphate aflibercept buffer, pH 6.2 solution aflibercept 5 mg/ml F6H8 non-aqueous sucrose suspension of aflibercept protein powder preparation bevacizumab 5 mg/ml 10 mM sodium aqueous trehalose phosphate bevacizumab buffer, pH 6.2 solution bevacizumab 5 mg/ml F6H8 non-aqueous trehalose suspension of bevacizumab protein powder preparation

(101) At the end of the treatment period, the animals were sacrificed, and their corneas analysed for the presence of blood and lymphatic vessels.

Example 4: Morphological Determination of Hemeangiogenesis and Lymphangiogenesis

(102) The animal corneas were excised, rinsed in PBS and fixed in acetone for 30 min. After three additional washing steps in PBS and blocking with 2% BSA in PBS for 2 h the corneas were stained overnight at 4° C. with rabbit anti-mouse LYVE-1. On day two the tissue was washed, blocked and stained with FITC-conjugated rat anti-CD31 (Acris Antibodies GmbH, Hiddenhausen, Germany) antibody overnight at 4° C. After a last washing and blocking step on day three, a goat-anti-rabbit Cy3-conjugated secondary antibody was used. Isotype control was assured with an FITC-conjugated normal ratIgG2A for CD31/FITC and with a normal rabbit IgG (both Santa Cruz Biotechnology, Santa Cruz, CA, USA) for LYVE-1.

(103) Double stained whole mounts were analysed with a fluorescence microscope (BX51, Olympus Optical Co., Hamburg, Germany) and digital grey value pictures were taken with a 12-bit monochrome CCD camera (F-View II, Soft Imaging System, Munster, Germany) at a resolution of 1376×1023 pixel. For the FITC stained blood vessels an HQ-FITC selectiv filterset (Exciter: HQ 480/40; Emitter: HQ 527/30; AHF Analysentechnik AG, Tübingen, Germany) was used. For the Cy3 stained lymphatic vessels the U-MWG2 mirror unit (Excitation filter: 510-550 nm; Emission filter: 590 nm; Dichromatic mirror: 570 nm; Olympus, Hamburg Germany) was used. Each whole mount picture was assembled out of 9 pictures taken at 100× magnification. The inhibition of blood vessel growth or lymph vessel growth are depicted in FIGS. 1 to 4.

Example 5: Early Effect

(104) Six- to eight-week-old female BALB/c mice were used for the model of suture-induced inflammatory neovascularization. Three 11-0 nylon sutures were placed intrastromally on the right eye with two stromal incursions extending over 120 degrees of corneal circumference each. The outer point of suture placement was chosen near the limbus and the inner suture point was placed near the center of cornea equidistant from the limbus to obtain standardized angiogenic responses. Mice were divided into 2 groups (n=10): aflibercept suspended in semifluorinated alkanes

(Afli/F6H8); aflibercept dissolved in Phosphate (Afli). Right after the suturing, drops were applied three times per day. Five and ten days after topical administration, corneas were harvested and stained with LYVE-1 and CD31 to quantify corneal hem- and lymphangiogenesis morphometric. FIG. 5 shows the early effect (5 days, 10 days) of inhibition of growth of both blood vessels and lymphatic vessels upon treatment with an ophthalmic composition comprising protein powder preparation of aflibercept suspended in F6H8 (Afli/F6H8). This effect is comparable to the aqueous aflibercept formulation (Afli).

Example 6: Low Dose Application

(105) The procedure is similar to Example 5, except for the treatment regime. Drops were applied only one time per day (3 μ l) and corneas were harvested 5 days after topical application. FIG. 6 shows that the effect of inhibition of growth of both blood vessels and lymphatic vessels can be achieved also with a reduced dosage. Herein, treatment with an ophthalmic composition comprising protein powder preparation of aflibercept suspended in F6H8 (Afli/F6H8) shows a comparable effect when compared to the aqueous aflibercept formulation (Afli).

Example 7: Effect of Inflammatory Cells

(106) Six- to eight-week-old female BALB/c mice were used for the model of suture-induced inflammatory neovascularization. Three 11-0 nylon sutures were placed intrastromally on the right eye with two stromal incursions extending over 120 degrees of corneal circumference each. The outer point of suture placement was chosen near the limbus and the inner suture point was placed near the center of cornea equidistant from the limbus to obtain standardized angiogenic responses. Mice were divided into 2 groups (n=10): aflibercept suspended in semifluorinated alkanes (Afli/F6H8); aflibercept dissolved in Phosphate (Afli). Right after the suturing, drops were applied three times per day. Three days after topical administration, corneas were harvested and stained with LYVE-1 and CD45 to quantify CD45+ cells and LYVE-1 macrophages. FIG. 7 shows the infiltration of inflammatory cells such as CD45+ cells and LYVE-1 macrophages at the early phase after 3 days. It appears that the ophthalmic composition comprising protein powder preparation of aflibercept suspended in F6H8 (Afli/F6H8) relates to slightly less infiltration of inflammatory cells when compared to the aqueous aflibercept formulation (Afli).

(107) FIG. 1 shows the inhibition of blood vessel growth (hemeangiogenesis) in the suture-induced inflammatory neovascularization mouse model utilizing the anti-VEGF protein Aflibercept Both, the diluted commercial aqueous composition ("Afli"; 5 mg/ml aflibercept dissolved in 10 mM sodium phosphate buffer, pH 6.2) as well as the non-aqueous ophthalmic suspension of protein powder preparations (comprising aflibercept) in F6H8 ("Afli/F6H8"; 5 mg/ml aflibercept suspended in F6H8) significantly inhibited the ingrowth of blood vessels ($p < 0.05$).

(108) FIG. 2 shows the inhibition of lymph vessel growth (lymphangiogenesis) in the suture-induced inflammatory neovascularization mouse model utilizing the anti-VEGF protein Aflibercept Both, the diluted commercial aqueous composition ("Afli"; 5 mg/ml aflibercept dissolved in 10 mM sodium phosphate buffer, pH 6.2) as well as the non-aqueous ophthalmic suspensions of protein powder preparations (comprising aflibercept) in F6H8 ("Afli/F6H8"; 5 mg/ml aflibercept, suspension in F6H8) significantly inhibited the ingrowth of lymph vessels ($p < 0.01$).

(109) FIG. 3 shows the inhibition of blood vessel growth (hemeangiogenesis) in the suture-induced inflammatory neovascularization mouse model utilizing the anti-VEGF protein avastin. Both, the diluted commercial aqueous composition ("Bevacizumab"; 5 mg/ml dissolved in 10 mM sodium phosphate buffer, pH 6.2) as well as the non-aqueous ophthalmic suspension of protein powder preparations (comprising bevacizumab) in F6H8 ("Be/F6H8"; 5 mg/ml bevacizumab; suspension in F6H8) inhibited the ingrowth of blood vessels.

(110) FIG. 4 shows inhibition of lymph vessel growth (lymphangiogenesis) in the suture-induced inflammatory neovascularization mouse model utilizing the anti-VEGF protein avastin. Both, the diluted commercial aqueous composition ("Bevacizumab"; 5 mg/ml dissolved in 10 mM sodium phosphate buffer, pH 6.2) as well as the non-aqueous ophthalmic suspension of protein powder

preparations (comprising bevacizumab) in F6H8 (“Be/F6H8”; 5 mg/ml bevacizumab; suspension in F6H8) inhibited the ingrowth of lymph vessels.

Claims

1. A method of treatment of corneal neovascularization and/or corneal angiogenesis, comprising administering a non-aqueous topical ophthalmic composition topically to the eye or an ophthalmic tissue of a subject in need thereof, wherein the composition comprises particles of a protein powder preparation suspended in a liquid vehicle comprising a semifluorinated alkane, wherein the particles of the protein powder preparation comprise aflibercept.
 2. The method according to claim 1, wherein the particles of the protein powder preparation are obtained by spray-drying or lyophilization of an aqueous composition comprising the aflibercept.
 3. The method according to claim 1, wherein the particles of the protein powder preparation further comprise one or more protein stabilizing agents selected from polyols, sugars, amino acids, amines and salting out salts, polymers, surfactants, and arginine.
 4. The method according to claim 1, wherein the concentration of the aflibercept in the composition is between about 1 to 60 mg/ml, or about 5 mg/ml.
 5. The method according to claim 1, wherein the size of 90% of the suspended particles of the protein powder preparation is between 1 to 100 μm , as measured by laser diffraction.
 6. The method according to claim 1, wherein the semifluorinated alkane is selected from F4H4, F4H5, F4H6, F6H4, F6H6, F6H8, and F8H8.
 7. The method according to claim 1, wherein the method is for the treatment of ocular neovascularization.
 8. The method according to claim 1, wherein the method provides inhibition of growth of blood vessels and/or lymphatic vessels.
 9. The method according to claim 1, wherein the method is for the treatment of and/or corneal angiogenesis.
 10. The method according to claim 1, wherein the subject is undergoing corneal transplantation.
 11. The method according to claim 1, wherein the particles of the protein powder preparation further comprise one or more protein stabilizing agents selected from sugars.
 12. The method according to claim 11, wherein the sugar is sucrose.
 13. The method according to claim 1, wherein the semifluorinated alkane is F6H8.
 14. The method according to claim 1, wherein the concentration of the aflibercept in the composition is between about 1 to 25 mg/ml, the semifluorinated alkane is F6H8, and wherein the particles of the protein powder preparation further comprise one or more protein stabilizing agents comprising sucrose.
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